

4	(a) work done in bringing unit positive charge from infinity (to that point)	M1	A1	[2]
	(b) (i) field strength is potential gradient	B1		[1]
	(ii) field strength proportional to force (on particle Q) potential gradient proportional to gradient of (potential energy) graph so force is proportional to the gradient of the graph	B1	B1	A0
				[2]

Page 3	Mark Scheme: Teachers' version	Syllabus	Paper
	GCE AS/A LEVEL – May/June 2011	9702	41

(c) energy = $5.1 \times 1.6 \times 10^{-19}$ (J) C1
 potential energy = $Q_1 Q_2 / 4\pi\epsilon_0 r$ C1
 $5.1 \times 1.6 \times 10^{-19} = (1.6 \times 10^{-19})^2 / 4\pi \times 8.85 \times 10^{-12} \times r$ C1
 $r = 2.8 \times 10^{-10}$ m A1 [4]

(d) (i) work is got out as x decreases M1
 so opposite sign A1 [2]

(ii) energy would be doubled B1
 gradient would be increased B1 [2]

5 (a) region (of space) where there is a force M1